



Animal Collection Report 2024



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**Pandemic literacy and viral zoonotic spillover risk at the
frontline of disease emergence in Southeast Asia to improve
pandemic preparedness**

PANDASIA

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1	CU	Chulalongkorn University	Thailand	WP1 lead
2	MU	Mahidol University	Thailand	WP2 lead
3	KKU	Khon Kaen University	Thailand	WP3 lead
4	UMU	Umeå University	Sweden	WP4 lead
5	UKHD	Universitätsklinikum Heidelberg University	Germany	WP5 lead
6	SUPA71	SUPA71 Co., Ltd.	Thailand	WP6 lead
7	NMBU	Norwegian University of Life Sciences	Norway	The Coordinator (WP7)
8	QMUL	Queen Mary University of London	UK	WP1-Partner
9	NVI	Norwegian Veterinary Institute	Norway	WP4-Partner
10	IZW	Leibniz Institute for Zoo and Wildlife Research	Germany	WP2-Partner
11	NOVA	UNIVERSIDADE NOVA DE LISBOA	Portugal	WP9 lead



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1. Executive Summary

This study presents a comprehensive investigation of wildlife and domestic animal populations across Chanthaburi and Chiang Rai provinces in 2024. A total of 1,056 animals were collected, including 412 bats, 304 rats, 24 secondary wildlife species, and 316 domestic animals. The study covered various habitats, such as bat caves, forests, orchards, dump site, households, and markets, to assess animal abundance, species diversity and potential interactions among wildlife (bats and rats), secondary wildlife, and domestic animals. Notably, bat diversity was higher in Chiang Rai, while household environments had the highest rat populations. Secondary wildlife was caught from multiple habitats, including agro-forests and dumpsites, and domestic animals were primarily free-roaming, increasing their potential risk of pathogen exposure.

Biological sample collection was systematically conducted across all animal groups. A total of 1,395 biological specimens were obtained, including oral, rectal, and nasal swabs, blood, urine, feces, and organ tissues. Specific sampling protocols were followed to ensure optimal sample quality for further analysis. The collected samples were preserved and transported under controlled conditions to facilitate subsequent laboratory analysis for further pathogen detection.

These animal results provide population size, abundance, and distribution of wildlife (bats and rats), secondary wildlife, and domestic animals across different provinces, habitats, and seasons. This information is crucial for modeling host population dynamics and assessing the potential risk of zoonotic pathogen transmission. Further longitudinal investigation in the second year can provide additional data and insights into the ecological factors influencing characteristics of animal populations.

2. Introduction & Background

The recent pandemics and epidemics of zoonotic diseases, such as COVID-19 and monkeypox, highlight the urgent need to better understand the factors and processes driving their emergence, as well as strategies for effective prevention. Zoonoses, which arise from animal-to-human spillover events, account for 60% of all emerging infectious diseases (EIDs) in humans (Taylor et al., 2001; Jones et al., 2008). Spillover is a complex phenomenon involving intricate biological, ecological, environmental, and human-related interactions. Regions with tropical forests, undergoing significant land-use changes and hosting rich biodiversity, are particularly prone to zoonotic disease emergence, including Southeast Asia and the Greater Mekong Subregion (GMS).

The GMS is recognized as a global hotspot for biodiversity, harboring a wealth of yet-undiscovered viral diversity and a heightened risk of EIDs. These risks are especially pronounced

in areas where wildlife-rich habitats intersect with human activities (Jones et al., 2008). The region is also home to a high diversity of rhinolophid bats, in which several SARS-CoV-2-related coronaviruses have been identified (Zhou et al., 2021). In fact, the closest known relatives to the SARS-CoV-2 virus responsible for COVID-19 have been detected in cave-dwelling bats inhabiting the limestone karst environments typical of this area (Temmam et al., 2022).

Further compounding these risks, infrastructure projects such as the high-speed rail line connecting Yunnan Province in China to Thailand via Laos, as part of the Kunming–Singapore railway under the Belt and Road Initiative, are expected to increase human mobility. This enhanced connectivity may facilitate the spread of zoonotic pathogens. Consequently, the GMS is a region of significant concern for the emergence of future pandemics, which could have far-reaching global impacts.

A recent report by the Intergovernmental Platform on Biodiversity and Ecosystem Services (IPBES, 2020) identified understanding viral diversity in wildlife with potential for future emergence as a critical knowledge gap. Effective pandemic preparedness requires proactive measures to mitigate spillover events and disease emergence. Addressing these risks necessitates local-level insights into circulating pathogens, particularly viruses, and their interactions with ecological systems and human activities. Developing targeted prevention strategies relies on a comprehensive understanding of these dynamics.

Spillover risks are amplified by environmental and ecosystem degradation, agricultural intensification, urbanization, and the loss of biodiversity (Glidden et al., 2021). These factors are driven by underlying social and economic pressures, such as food insecurity and financial instability, which disrupt ecosystems and promote wildlife consumption and trade. Local communities play a crucial role at the interface of animals, humans, and the environment, shaping the dynamics of disease emergence and transmission. Understanding the social, behavioral, and structural risks, along with the drivers and barriers influencing spillover at the local level, is essential.

The significance and impact of these factors vary based on cultural practices, community norms, and literacy levels. Consequently, research that targets local communities within identified high-risk areas can provide more tailored and actionable insights. Adopting a holistic approach that integrates these diverse factors into a unified system allows for a deeper focus on human and social drivers of spillover. This understanding creates opportunities to implement changes through effective communication and action.

Rodents, bats, and birds are highly diverse animal groups, with many species well-adapted to human-altered environments. These animals are recognized as significant reservoirs of various zoonotic parasites and pathogens. Additionally, they play critical roles in ecosystems that directly influence human health, economic stability, and food production. Their ecological contributions include seed dispersal, plant pollination, participation in food webs as both consumers and prey, and natural pest control.

In Thailand, murid rodents (rats and mice) and bats exhibit remarkable diversity, with 36 and 92 species documented, respectively (Lekagul and McNeely, 1977). From a public health perspective, rats and mice are prominent disease carriers and reservoirs of numerous zoonotic pathogens. Meerburg et al. (2009) conducted a comprehensive review highlighting over 60 zoonotic pathogens associated with murid rodents, including microparasites (e.g., viruses, bacteria, and protozoa) and macroparasites (e.g., helminths). For instance, rodent-borne viral diseases pose significant health risks: hantaviruses, transmitted through airborne exposure, can cause severe hemorrhagic renal and pulmonary syndromes (Jonsson et al., 2010). Another example is hepatitis E virus, a pathogen transmitted via the fecal-oral route, often through contaminated food from infected animals like rodents, wild boars, deer, or pigs, leading to foodborne illness in humans (Sridhar et al., 2021).

Bats, on the other hand, are widely known as reservoirs of emerging and re-emerging viral diseases. Over 80 different viruses have been identified in bats, many of which are associated with human diseases (Melaun et al., 2014). In Southeast Asia, large fruit bats (genus *Pteropus*) harbor several highly pathogenic viruses, such as Nipah virus, Hendra virus, and Australian bat lyssavirus, all of which can be fatal to humans (Mackenzie et al., 2003; Gay, 2014). Additionally, insectivorous bats have been linked to other deadly viruses, including rabies and severe acute respiratory syndrome coronaviruses (SARS-CoVs) (Calisher et al., 2006; Cargnelutti et al., 2021). In this study, murine rodents, bats, other wildlife, and domestic animals (e.g., dogs and cats) were trapped or approached to collect biological samples, including oro-nasal swabs, rectal swabs, and blood, for pathogen surveys of viruses circulating in animal hosts. Ecological data were collected to assess animal distribution and abundance, focusing on spatial patterns to identify animal-human interfaces and evaluate the risk of pathogen spillover events.

The combined data on animal hosts harboring high-risk spillover viruses, along with findings from social science research (not detailed in this protocol), will be used to inform the development and delivery of a pandemic preparedness and prevention literacy intervention. The overarching project, “PANDASIA,” will evaluate the effectiveness of these interventions and their implementation to guide decision-making and assess the potential for scaling and adaptation in other contexts.

The overall impact of this project is expected to extend beyond local communities, contributing to a deeper understanding of the pathogen spillover system and supporting the development of sustainable strategies for pandemic preparedness and prevention.

3. Objectives

The animal study belongs to a part of the main PANDASIA project and focuses on identifying key wild vertebrate hosts (such as rodents and insectivorous bats) that pose a spillover risk in two selected locations: Chiang Rai and Chanthaburi Provinces, Thailand.

The specific objectives are to discover the presence, abundance, and diversity of wild and domestic animals in study sites to map spatial and temporal changes in land use change and cover, population size, and climate.

This study primarily targets bats and rodents due to their abundance and potential role in spillover events. However, other wildlife and domestic animals (e.g., squirrels, dogs, and cats) may also be important as intermediary hosts. Animals were trapped or approached to collect biological samples.

In addition to biological sampling, detailed data regarding each animal's health status, prior medical history, veterinary care practices, behavioral patterns, and living conditions were systematically recorded using standardized data collection forms.

4. Methods

4.1. Study sites

The project activities conducted at selected communities in Chiang Rai and Chanthaburi Provinces. These provinces reflect a diversity of biological, ecological, societal contexts and border areas of Greater Mekong Subregion (GMS). Each includes a provincial city, district towns, sub-district towns and many villages, nature areas, contiguous forest areas, national parks, scenic sites, which are often frequented by tourists and other visitors. Karstic limestone caves known for housing bat colonies, from which samples can be obtained, are also common in these areas. Animal investigation was conducted across four districts in two provinces. In Chiang Rai Province, the study sites included Wiang Kaen and Mae Fah Luang districts, while in Chanthaburi Province, the investigations were carried out in Pong Nam Ron and Soi Dao districts.

In each district, various study sites were investigated to represent the potential of human-wildlife interface and spillover risk: natural forests, rural village settings, and district towns. The study sites were categorized based on specific habitat types to capture a comprehensive understanding of ecological and anthropogenic factors influencing spillover dynamics. These habitat types were defined as follows: forest (F), bat cave (B), orchard (Q), perennial crop area (P), field crop area (C), paddy field (D), solid waste site (S), temple (T), and community/household areas (H). This classification allows systematic data collection across diverse landscapes and interface settings. For bats, investigations were conducted in two primary habitats: bat caves and fruit orchards, chosen based on their ecological significance for bat populations. For rats, live traps were set across a diverse range of habitats, including forests, orchards, field crop areas, paddy fields, solid

waste sites, and household areas, to capture a broad representation of species inhabiting different environments. Secondary wildlife was obtained from markets and opportunistically from locations where bats and rats were captured. For domestic animals, including dogs and cats, sampling was restricted to household areas, as these animals are closely associated with human dwellings.

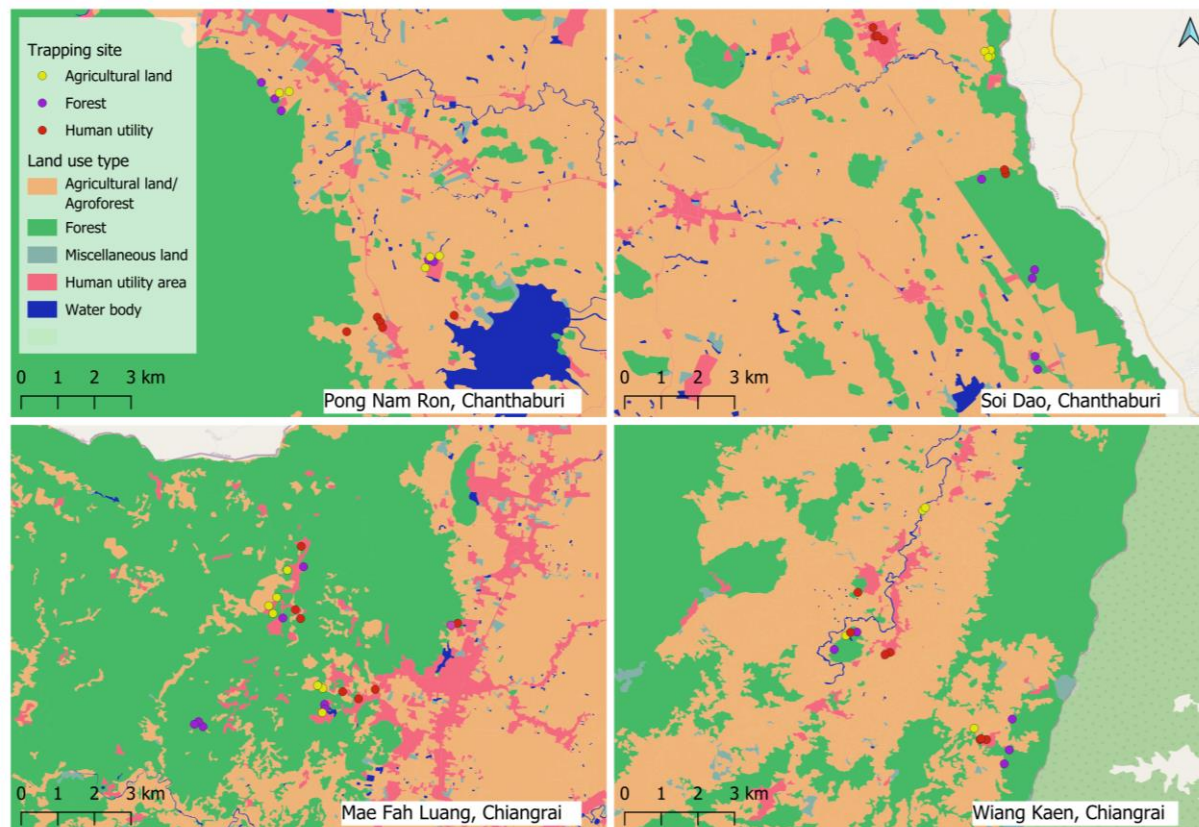


Figure 4.1-1 Animal study sites across various landuse types i.e. agriculture land, forest, and human utility in each 2 locations of Chanthaburi and Chaing Rai.

4.2. Animal trapping or approach

Animal sample collection was conducted in Chanthaburi and Chiang Rai, across both dry and wet seasons in 2024. In Chanthaburi province, the survey took place in Soi Dao and Pong Nam Ron districts, with the dry season collection occurring from April 2 to 8, 2024, and the wet season collection from July 8 to 15, 2024. In Chiang Rai province, the survey was conducted in Wiang Kaen and Mae Fah Luang districts, with the dry season collection from April 21 to 29, 2024, and the wet season collection from October 18 to 26, 2024. Four districts as the study site in Thailand was demonstrate in Figure 5.1.1

4.3. Bats capture and handling

Insectivorous bats were captured using four-bank harp traps and monofilament mist nets (3 × 9 m with a mesh size of 16 × 16 mm) placed at and near cave entrances. For the capture of frugivorous bats, monofilament mist nets were deployed along flight pathways in fruit orchards. Traps were actively monitored to ensure animal welfare, with checks conducted every 30 minutes between 18:00 and 21:00. At least one researcher remained present to supervise and respond promptly to any entanglement.

Captured bats were immediately removed from the traps to minimize injury, strangulation, or stress. Each bat was placed in an individual cloth bag by a well-trained or experienced handler to ensure safe removal without causing harm to the delicate wing bones and patagia. The bats were then weighed using either a 10 g or 100 g Pesola Spring balance, with a precision of 0.5 g. External morphological measurements were taken using dial calipers to the nearest 0.5 mm, following standardized protocols outlined by Bates and Harrison (1997). Species identification was conducted based on external characteristics, following the guideline of the book of Bats of Thailand: for field identification (Duengkae, 2007)

The handling of bats adhered to the guidelines set forth by the American Society of Mammalogists (Sikes, 2016). All procedures, including biological specimen collection (detailed in the subsequent section), were conducted in the field to ensure minimal disturbance to the animals. Following data collection, each bat was released at the exact location where it had been captured to maintain ecological integrity and natural behavioral patterns.

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Figure 4.3-1 four-bank harp traps.



Figure 4.3-2 mist net traps at cave entrance.

4.4. Rodent Capture and Handling

Murine rodents were trapped alive using 10 mesh wire traps, with one trap placed 15 m apart along 150 m sampling lines in each study environment. Fresh corn was used as bait to attract rodents. Traps were set overnight from 18:00 to 06:00 for four consecutive nights. To ensure minimal stress to the captured animals, each trap containing a rodent was covered with a dark-colored cloth bag, allowing for adequate ventilation while also reducing visual stress. The traps were then placed on plastic trays to facilitate the collection of fecal samples before specimen processing.

Captured rodents were transported to a designated setup area for biological specimen collection. To ensure safe handling, the rodents were anesthetized using isoflurane inhalation. The anesthesia was conducted in open-air areas, away from other captured animals, to prevent cross-contamination and minimize stress. The entire trap, with the rodent inside, was placed into a transparent, heavy-duty plastic bag (60-liter capacity) containing 5% isoflurane. Approximately 12 ml of isoflurane was soaked into cotton wool and placed within the bag to induce anesthesia. The rodents were closely monitored throughout the process to ensure their safety and prevent overexposure.

Species identification was conducted based on external morphology, following the taxonomic keys of Lekagul and McNeely (1977) and Aplin et al. (2003). Handling protocols adhered to the guidelines established by the American Society of Mammalogists (Sikes, 2016). To facilitate individual identification, rodents were tagged using ear punches, following standardized marking techniques appropriate for small mammal wildlife (Silvy et al, 2012).

Each rodent was weighed using either a 10 g or 100 g Pesola Spring balance, with a precision of 0.5 g. External morphological measurements were taken using dial calipers to the nearest 0.5 mm, following standardized protocols outlined in previous rodent studies. Additional biological data, including body condition, reproductive status, and any visible abnormalities, were also recorded.

For anesthetic recovery, rodents were returned to their traps and placed in a warm, clean, and dry environment away from other captured animals. They were continuously monitored until they regained an upright posture and exhibited normal locomotion within the trap. Once fully recovered, all rodents were released at the exact location where they had been captured to minimize disruption to their natural behavior and home ranges. Animal trapping sessions were performed and finished all the processes on a day-by-day basis. Animals were released at the area of capture immediately after the above measurements were completed. Any re-trapped animal was only recorded for recapturing and not subjected for body measurement or specimen collection again.

4.4.1. Secondary wildlife

The secondary wildlife e.g. squirrel, treeshews, civets and wild birds were captured or obtained from various habitats and techniques.

Small mammals bycatch, including squirrels and tree shrews, were captured and handled following the same protocol of murine rodent capture described as the previous topic by specifically adapted for these species. Capture methods utilized baited live traps placed in

appropriate microhabitats to maximize capture efficiency while minimizing stress. Traps were checked at regular intervals to ensure animal welfare. Species identification was conducted based on external morphology, using taxonomic keys provided by Lekagul and McNeely (1977) and Aplin et al. (2003). All handling procedures followed the ethical guidelines established by the American Society of Mammalogists (Sikes, 2016). For individual identification, ear punches were applied following standardized marking techniques suitable for small mammal wildlife (Silvy et al., 2012). Each animal was weighed using either a 10 g or 100 g Pesola Spring balance, with a precision of 0.5 g. External morphological measurements were recorded using dial calipers to the nearest 0.5 mm. Additionally, habitat characteristics at the capture site and biological data, including body condition, reproductive status, and any visible abnormalities, were documented for ecological and health assessments.

For medium-sized wild mammals, carcasses or heads of animals were obtained from bush meat markets, where they were purchased for the purpose of biological sample collection and further analysis. The carcasses were carefully transported to designated study areas, ensuring adherence to strict biosafety protocols to minimize the risk of contamination and to prevent exposure to any hazardous materials during handling. Upon arrival at the study site, the animals were handled for collection of biological samples, which included tissue samples, blood, and swabs from various anatomical regions, as well as internal organs when possible. Species identification was conducted based on morphological characteristics, using taxonomic keys and guidelines specific to Southeast Asian mammals (Francis, 2019). In addition to species identification, detailed ecological and biological data were recorded, including information on the condition of the carcasses, potential signs of disease or infection, and any visible abnormalities. This additional data provided insights into the health status of the wildlife population and contributed to the overall risk assessment for pathogen circulation and spillover potential.



Figure 4.4.1-1 An Indochinese ground squirrel (*Menetes berdmorei*) as secondary wildlife was live-trapped

Wild birds were occasionally captured as bycatch in the mist nets set up during bat capturing activities, as the two species often occupy overlapping habitats. The mist nets were strategically positioned in areas frequented by both bats and birds to maximize capture success. Upon capture, the birds were carefully removed from the nets by trained staff to prevent injury or stress.

Physical restraint techniques

were employed to ensure that the birds remained calm and were safely handled for biological sample collection and data recording. Biological samples and detailed data on each bird, such as health status, behavior, age, sex, and any visible signs of disease or injury, were recorded. This comprehensive data allowed for the assessment of pathogen circulation, potential zoonotic risks, and the general health of bird populations in the study area. Species identification was carried

out according to the guidelines of the bird guide of Thailand (Lekagul, 1974), using external morphological features such as plumage patterns, size, and bill structure. After data collection and sample acquisition were completed, all birds were released at the exact locations where they were captured to minimize any disruption to their natural behavior and home ranges.

4.4.2. Domestic animal restraint and handling

Domestic animal surveillance primarily focused on dogs and cats, as these species are commonly found in close association with human settlements, particularly within communities and household environments. To facilitate access to these animals and ensure ethical and effective sample collection, collaboration was established with government officials from the Department of Livestock Development and village health volunteers.

The sampling of domestic animals was conducted during routine veterinary and public health activities, such as government-organized rabies vaccination campaigns and sterilization programs. These events provided an opportunity to engage with pet owners and obtain biological samples. Dogs and cats were approached systematically, and efforts were made to include a diverse range of individuals from different household environments to ensure a representative dataset. For biological sample collection, each animal was assessed to determine the appropriate handling method. Physical restraint was applied in cases where the animal remained calm and cooperative. All procedures were performed under the supervision of licensed veterinarians to ensure animal welfare and adherence to ethical guidelines. A comprehensive set of biological samples was collected from each individual, including nasal swabs, oral swabs, rectal swabs, and blood samples. In addition to biological sampling, detailed data regarding each animal's health status, prior medical history, veterinary care practices, behavioral patterns, and living conditions were systematically recorded using standardized data collection forms.



Figure 4.4.2-1 Domestic animal restraint and biological sample collection.

4.5. Data and biological samples collection

4.5.1. Bats data and biological sample collection

Bat data, including the date and time of collection, weather conditions, habitat type, species identification, age, reproductive status, biometric measurements, and other relevant observations, were systematically recorded using a designated data collection form for comprehensive analysis.

For biological specimen collection from bats, trained staff carefully performed oropharyngeal and rectal swabbing under proper restraint, ensuring minimal stress to the animals. Sterile polyester-tipped swab applicators were used to collect secretions for pathogen detection. Given the small body size of bats, swabs measuring 1.981 millimeters in diameter with polyester tips and aluminum shafts (Puritan®, USA) were used for rectal and oral sampling. For medium-sized bats, flock-tipped swabs with a 3.556-millimeter diameter and polystyrene handles (Puritan®, USA) were also utilized for oral sample collection. To enhance pathogen recovery and increase the sensitivity of detection, two swabs were used per sample collection. Both swabs were then placed in the same collection tube to maximize sample yield. The collected swab samples were preserved in 0.2 ml of AVL buffer, a lysis buffer used in nucleic acid extraction protocols (Qiagen®). In addition to swab collection, urine and fecal samples were opportunistically collected from naturally excreted materials without any forced extraction. Urine samples were absorbed using flock-tipped swabs to optimize sample retention and were preserved in AVL buffer as the transport medium. Fecal samples, with an estimated weight of 10 milligrams, were directly placed into the preservative media.

All biological specimens were stored at 4°C and transported under controlled conditions, i.e., keeping the cold chain, to the laboratory at the Faculty of Medicine, Khon Kaen University within 72 hours of collection.

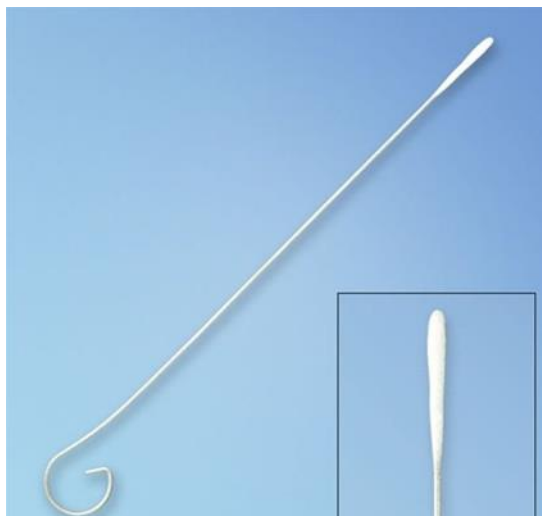


Figure 4.5.1-1 the polyester tips and aluminum shafts swab applicators for bats and rats.

4.5.2. Rat data and biological sample collection

Data on rats, including the date and time of collection, habitat type, species identification, age, sex, biometric measurements, and additional observations, were systematically documented using the designated data collection form.

For biological specimen collection, oropharyngeal and rectal swabs were obtained under controlled anesthesia to ensure both the safety of the animals and the reliability of sample collection. Sterile polyester-tipped swab applicators were used to collect secretions from oropharyngeal and rectal of animals for pathogen detection. Due to the small size of the rat's rectum, rectal swabs were collected using polyester-tipped swabs with a 1.981-millimeter diameter (Puritan®, USA), whereas oral swabs were taken using flock-tipped swabs with a 3.556-millimeter diameter (Puritan®, USA). To maximize sample yield, two swabs were used per collection and placed in the same collection tube. Swab samples were preserved in 0.2 ml of AVL buffer, a lysis buffer for nucleic acid extraction (Qiagen®). In addition to swab sampling, urine and fecal samples were opportunistically collected whenever naturally excreted by the animals. Urine was absorbed using flock-tipped swabs to maximize retention, while fecal samples, approximately 10 mg in weight, were directly transferred into a preservative solution for further laboratory processing. The whole blood samples were collected via facial vein or submandibular venipuncture using a sterile 1.8-mm (22G) lancet. A volume of 100–200 µl of blood was drawn and transferred into an EDTA-treated collection tube. This method has been recognized as a reliable and field-appropriate technique for small mammal blood collection (Berl et al., 2017). All biological specimens were stored at 4°C during fieldwork and transported under temperature-controlled conditions to the designated laboratory at the Faculty of Medicine, Khon Kaen University within 72 hours of collection.



Figure 4.5.2-1 Oral swab sample collection from murine rodent

4.5.3. Secondary wildlife data and biological sample collection

Data on secondary wildlife, including the date and time of collection, habitat type, species identification, and additional ecological observations were documented using the designated data collection form.

The collection of biological samples including oropharyngeal, rectal swab, urine and fecal samples from small mammals followed protocols similar to those used for rats. The blood samples of small mammals were collected via saphenous venipuncture by 23G or 24G needle. A volume of 100–200 µl of blood was drawn and transferred into an EDTA-treated collection tube.

Wild birds were captured opportunistically using mist nets during bat sampling efforts. Birds were carefully handled using physical restraint by trained personnel to minimize stress and injury. Oropharyngeal and cloacal swabs were collected using sterile polyester-tipped applicators and immediately placed in 0.2 ml of AVL buffer to preserve viral RNA integrity. Blood samples were collected via brachial wing venipuncture using a 23G or 24G needle. Approximately 100–200 µl of blood was drawn and transferred into an EDTA-treated collection tube.

For medium-sized wild mammals, fresh whole carcasses were bought from the bush meat markets and then necropsy to collect the vital organ tissue including lung, liver, spleen kidney and intestine under the sterile technique. The tissue sample was kept into the sterile tube.

For medium-sized wild mammals, fresh whole carcasses were acquired from bushmeat markets. Necropsies were performed and vital organ tissues—including the lungs, liver, spleen, kidneys, and intestines—were aseptically excised and placed into sterile tubes. Tissue samples were preserved under appropriate.

All biological specimens were temporarily stored at 4°C in field conditions and transported under controlled temperatures to the designated laboratory at the Faculty of Medicine, Khon Kaen University within 72 hours of collection.

4.5.4. Domestic animal data and biological sample collection

Data of dogs, cats and other domestic animals including animal name, owner name, owner address, age, sex weight, body condition score, appearance clinical signs, living behavior, and other observational remarked were collected to the designated record form.

Biological samples from domestic animal including oral swab, rectal swab, nasal swab and blood samples were conducted under the animal owner permission. Sample collection was conducted under the physical restraint by trained staff with animal owners.

Oropharyngeal swabs and rectal swabs were collected using sterile flocked swab specimen collection devices with flock-tipped swabs with a 3.556-millimeter diameter (Puritan®, USA) for pathogen detection. Swab samples were preserved in 0.5 ml of AVL buffer, a lysis buffer for nucleic acid extraction (Qiagen®). The whole blood samples were collected via cephalic vein or saphenous vein using a sterile 23G to 21G needle. A volume of 1–3 milliliter of blood was drawn and transferred into an EDTA-treated collection tube. The whole blood was submitted to centrifugation at 1000 g for 10 minutes to separate the plasma for further antibody detection.

The remaining red blood cell samples and separated plasma samples were separately transferred to a sterile tube of 1.5 ml for further pathogen infection analysis.

All biological specimens were stored at 4°C during fieldwork and transported under temperature-controlled conditions to the designated laboratory at the Faculty of Medicine, Khon Kaen University within 72 hours of collection.

5. Results

5.1. All animals

A total of 1,056 animals were investigated during the study, comprising bats, rats, secondary wildlife, and domestic animals. Among them, 412 bats were sampled, with 243 collected in Chanthaburi (103 in the dry season and 140 in the wet season) and 169 in Chiang Rai (75 in the dry season and 94 in the wet season). A total of 304 rats were examined, including 130 from Chanthaburi (70 in the dry season and 60 in the wet season) and 174 from Chiang Rai (102 in the dry season and 72 in the wet season). Secondary wildlife, including squirrels, treeshrews, and wild birds, accounted for 24 individuals, with 12 collected in Chanthaburi (8 in the dry season and 4 in the wet season) and 9 in Chiang Rai (4 in the dry season and 5 in the wet season). Additionally, 316 domestic animals, primarily dogs and cats, were sampled, with 190 from Chanthaburi (87 in the dry season and 103 in the wet season) and 126 from Chiang Rai (74 in the dry season and 52 in the wet season). These findings are summarized in Figure 5.1.2

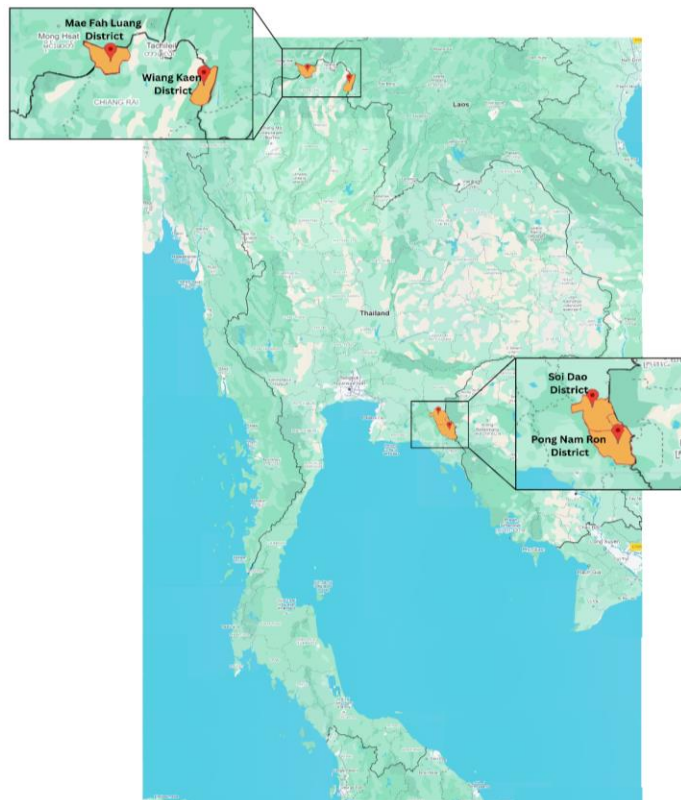


Figure 5.1-1 All four district study sites in Thailand.

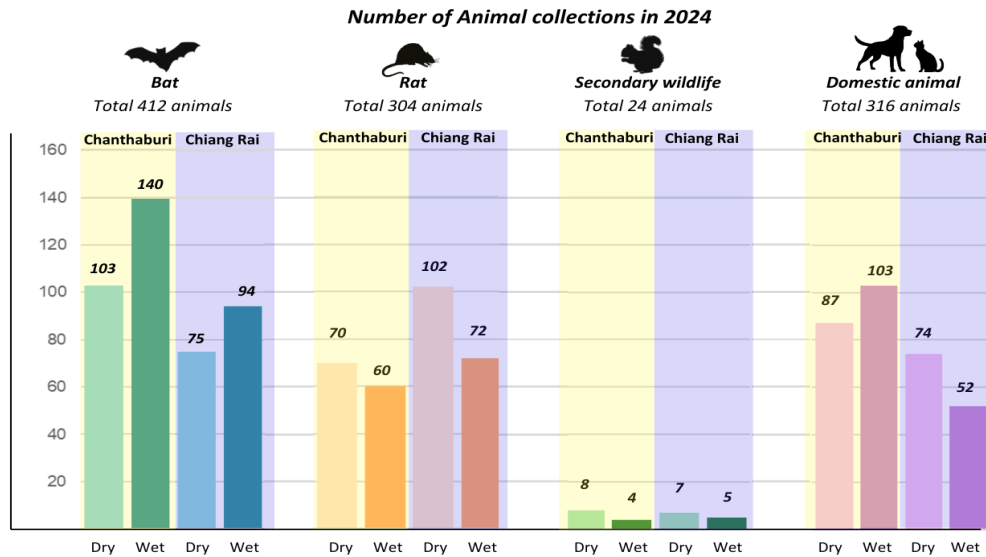


Figure 5.1-2 The number of bats, rats, secondary wildlife, and domestic animals collected in two provinces during two seasons in 2024 in Thailand.

5.2. Bats

A total of 412 bats were investigated across both study sites. In Chanthaburi Province, 243 bats were recorded, representing five families, eight genera, and 14 species. In Chiang Rai Province, 169 bats were identified, exhibiting greater taxonomic diversity with five families, ten genera, and 16 species. The detailed species composition in each province are presented in Table 5.2.1. and Figure 5.2.1

For trapping effort, a total of 63 mist nets and 35 harp traps were deployed for bat sampling in 2024. The selection and distribution of trapping methods were tailored based on habitat types and specific site conditions to maximize capture efficiency and minimize disturbance. The number and type of traps used varied across locations to accommodate differences in bat behavior and environmental characteristics. Detailed trapping efforts for each site are presented in Table 5.2-3.

The habitat distribution of bats varied across the study sites in Chanthaburi and Chiang Rai provinces. In Chanthaburi, the majority of bats were found in cave habitats (198 individuals), followed by forested areas (25 individuals) and orchards (20 individuals). In Chiang Rai, caves also served as the primary habitat, with 121 bats recorded, while agro-forests and village areas accounted for 39 and 9 individuals, respectively. These findings highlight the preference of bats for cave environments while also demonstrating their presence in diverse habitats. The detailed data are presented in Figure 5.2.2.

For biological sample collection, a total of 243 bats in Chanthaburi were sampled, yielding 220 rectal swabs, 37 fecal samples, and 45 urine sample. In Chiang Rai, 169 bats were investigated,

with biological samples comprising 79 rectal swabs, 25 fecal samples, and 39 urine samples. A detailed list of biological samples collected from each bat species in both provinces is presented in Table 5.2.2.

Table 5.2-1. Species composition of bats collected in Chanthaburi and Chaing Rai provinces in 2024.

Province and Season	Bats data Family	Genus	Species	No. of Animal		
				Dry	Wet	Total
Chanthaburi	Emballonuridae	<i>Taphozous</i>	<i>Taphozous melanopogon</i>	61	76	
	Hipposideridae	<i>Hipposideros</i>	<i>Hipposideros armiger</i>	11	14	
			<i>Hipposideros galeritus</i>	0	2	
			<i>Hipposideros gentilis</i>	3	4	
			<i>Hipposideros larvatus</i>	4	21	
	Rhinolophidae	<i>Rhinolophus</i>	<i>Rhinolophus pusillus</i>	6	1	
			<i>Rhinolophus shameli</i>	4	3	
			<i>Rhinolophus thomasi</i>	2	0	
			<i>Rhinolophus malayanus</i>	0	3	
	Pteropodidae	<i>Cynopterus</i>	<i>Cynopterus cf. brachyotis</i>	0	6	
		<i>Eonycteris</i>	<i>Eonycteris spelaea</i>	9	4	
		<i>Macroglossus</i>	<i>Macroglossus sobrinus</i>	0	3	
		<i>Rousettus</i>	<i>Rousettus leschenaultii</i>	1	3	
	Vespertilionidae	<i>Myotis</i>	<i>Myotis siligorensis</i>	2	0	
	Total			103	140	243
Chiang Rai	Hipposideridae	<i>Aselliscus</i>	<i>Aselliscus stolickanus</i>	29	53	
		<i>Hipposideros</i>	<i>Hipposideros armiger</i>	3	4	
			<i>Hipposideros larvatus</i>	3	1	
	Pteropodidae	<i>Cynopterus</i>	<i>Cynopterus sphinx</i>	0	16	
		<i>Eonycteris</i>	<i>Eonycteris spelaea</i>	1	0	
		<i>Macroglossus</i>	<i>Macroglossus sobrinus</i>	2	7	
		<i>Rousettus</i>	<i>Rousettus amplexicaudatus</i>	1	0	
			<i>Rousettus leschenaultii</i>	0	2	
	Megadermatidae	<i>Eudiscoderma</i>				
		<i>a</i>	<i>Lyroderma lyra</i>	0	1	
	Rhinolophidae	<i>Rhinolophus</i>	<i>Rhinolophus pusillus</i>	7	9	
			<i>Rhinolophus marshalli</i>	0	1	
			<i>Rhinolophus thailandensis</i>	5	0	
			<i>Rhinolophus thomasi</i>	21	0	
	Vespertilionidae	<i>Myotis</i>	<i>Myotis siligorensis</i>	1	0	
			<i>Myotis Sp.</i>	1	0	
		<i>Scotophilus</i>	<i>Scotophilus kuhlii</i>	1	0	
	Total			75	94	169
Summary			178	234	412	

Chanthaburi



Cynopterus cf. brachyotis



Eonycteris spelaea



Hipposideros armiger



Hipposideros galeritus



Hipposideros gentilis



Hipposideros larvatus



Macroglossus sobrinus



Myotis siligorensis



Rhinolophus malayanus



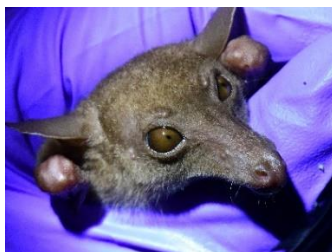
Rhinolophus pusillus



Rhinolophus shameli



Rhinolophus thomasi

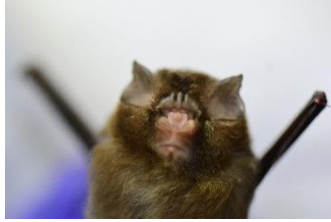


Rousettus leschenaultii



Taphozous melanopogon

Chiang Rai



Aselliscus stolickanus



Cynopterus sphinx



Eonycteris spelaea



Hipposideros armiger



Hipposideros larvatus



Macroglossus sobrinus



Myotis siligorensis



Myotis Sp.



Rhinolophus marshalli



Rhinolophus pusillus



Rhinolophus thailandensis



Rhinolophus thomasi



Rousettus amplexicaudatus



Rousettus leschenaultii



Scotophilus kuhli

Lyroderma lyra; unavailable picture

Table 5.2-2 Biological sample collection from bats in Chanthaburi and Chaing Rai in 2024.

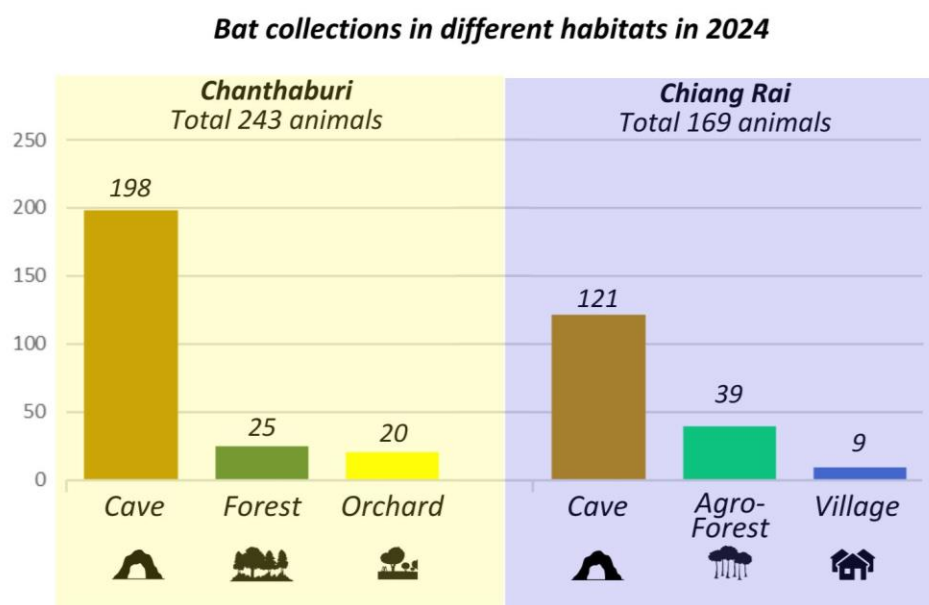


Figure 5.2-1 Bat collections in different three habitats in this study from Chanthaburi and Chiang Rai in 2024

Province	Species	No. of animal	Rectal swab	Feces	Urine
Chanthaburi	<i>Taphozous melanopogon</i>	137	132	9	13
	<i>Hipposideros armiger</i>	25	25	9	10
	<i>Hipposideros larvatus</i>	25	25	9	11
	<i>Eonycteris spelaea</i>	13	11	1	1
	<i>Hipposideros gentilis</i>	7	4	1	2
	<i>Rhinolophus pusillus</i>	7	1	1	2
	<i>Cynopterus cf. brachyotis</i>	6	6	0	0
	<i>Rhinolophus shameli</i>	4	3	1	0
	<i>Rousettus leschenaultii</i>	4	4	0	1
	<i>Rhinolophus malayanus</i>	3	0	2	2
	<i>Macroglossus sobrinus</i>	3	3	0	0
	<i>Rhinolophus shameli</i>	3	3	1	1
	<i>Hipposideros galeritus</i>	2	2	1	0
	<i>Myotis siligorensis</i>	2	0	1	1
	<i>Rhinolophus thomasi</i>	2	1	1	1
		243	220	37	45
Chiang Rai	<i>Aselliscus stolickanus</i>	82	17	8	17
	<i>Rhinolophus thomasi</i>	21	13	4	5

<i>Cynopterus sphinx</i>	16	16	3	1
<i>Rhinolophus pusillus</i>	16	2	4	4
<i>Macroglossus sobrinus</i>	9	9	1	3
<i>Hipposideros armiger</i>	7	6	2	3
<i>Hipposideros larvatus</i>	4	4	1	1
<i>Rhinolophus thailandensis</i>	4	3	1	1
<i>Rousettus leschenaultii</i>	2	2	0	1
<i>Eonycteris spelaea</i>	1	1	0	1
<i>Lyroderma lyra</i>	1	1	0	0
<i>Myotis siligorensis</i>	1	1	0	0
<i>Myotis Sp.</i>	1	0	0	1
<i>Rhinolophus marshalli</i>	1	1	1	1
<i>Rhinolophus pearsonii</i>	1	1	0	0
<i>Rousettus amplexicaudatus</i>	1	1	0	0
<i>Scotophilus kuhlii</i>	1	1	0	0
	169	79	25	39
Total	412	299	62	84

Table 5.2-3 Bat trap effort data in Chanthaburi and Chaing Rai in 2024.

			No. of Bat	No. of mist net	No. of harp trap
Dry season	Chanthaburi	Cave	81	9	0
		Forest	15	3	2
		Orchard	7	7	4
				19	6
	Chiang Rai	Cave	63	3	4
		Agro-forest	10	8	3
		Village	2	3	2
				14	9
Wet season	Chanthaburi	Cave	117	5	2
		Forest	10	6	4
		Orchard	13	6	4
				17	10
	Chiang Rai	Cave	58	3	4
		Agro-forest	29	6	4
		Village	7	4	2
				13	10

5.3. Rodents

A total of 304 rats were collected in 2024, with 130 individuals from Chanthaburi and 174 from Chiang Rai. All collected rats belonged to the family Muridae. In Chanthaburi, five genera with eight species were identified: *Rattus*, *Mus*, *Maxomys*, *Berylmys*, and *Bandicota*. In Chiang Rai, a higher diversity of seven genera with ten species was observed, including *Rattus*, *Mus*, *Maxomys*, *Berylmys*, *Niviventer*, *Hapalomys*, and *Leopoldamys*. Seasonally, 172 rats were captured during the dry season, while 132 were collected in the wet season. A detailed breakdown of rat species distribution across different seasons and study locations is provided in Table 5.3.1.

The habitat analysis in this study identified six distinct habitat types where rats were captured. Household areas were the most common habitat, yielding the highest number of rats across both seasons in 2024. In Chanthaburi, a total of 92 rats were obtained from households, followed by 14 from forests, 11 from orchards, 12 from dumpsites, and a single rat from a field crop. Similarly, in Chiang Rai, 117 rats were captured from households, 33 from forests, 5 from orchards, 9 from dumpsites, 9 from paddy fields, and 1 from a field crop. The detailed distribution of rat captures across different habitats is presented in Figure 5.3.1.

The habitat-wise species richness revealed notable differences in rat distribution across various environments. While the highest number of rats (209 individuals) was recorded in household habitats across both provinces, species richness within this habitat was relatively low, with only five identified species. Notably, 164 out of 209 individuals (approximately 78.47%) belonged to the genus *Rattus*, predominantly represented by *Rattus exulans* (156 individuals) and the *Rattus rattus* complex (48 individuals). In contrast, forest habitats exhibited the highest species richness, with eight distinct species: *Rattus rattus* complex, *Rattus andamanensis*, *Maxomys surifer*, *Niviventer fulvescens*, *Hapalomys delacouri*, *Leopoldamys edwardsi*, *Mus pahari*, and *Rattus exulans*. These findings suggest that while household environments support larger rat populations, forested areas harbor a greater variety of species, emphasizing the role of habitat complexity in shaping rodent biodiversity. The detailed distribution of rat species across habitats is presented in Table 5.3.2.

To analyze trends in rodent abundance across different land use types, rodents from sampling points were grouped to calculate the Relative Abundance Index (RAI), defined as the number of rodents per 100 trap-nights. The RAI was then compared across habitat types and seasons within each district. Rodents captured from field crops, paddy fields, and orchards were grouped together under agriculture, while those from households and dumpsites were categorized under utility. In Pong Nam Ron, the highest RAI was observed in forest areas during the dry season (4.67), whereas the lowest was in agriculture during the wet season (0.67). The utility land use type exhibited a higher RAI in the dry season (2.3) compared to the wet season (0.67). In Soi Dao, the highest RAI occurred in utility areas during the dry season (7.33), decreasing to 4 in the wet season. Forest and agriculture areas showed higher RAI in the dry season (3.33 each), while

agriculture had the lowest RAI in the wet season (0.67). In Mae Fah Luang, the highest RAI was found in forested areas during the dry season (7.62), followed by utility areas (5.56). In the wet season, the RAI was lower across all land use types, with the highest recorded in utility areas (2.78). In Wiang Kaen, the RAI in utility areas remained relatively consistent between seasons (4.84 in dry and 5 in wet), while forest areas had a slightly higher RAI in the dry season (4.46) than in the wet season (1.52). Agriculture had the lowest RAI across all sites and seasons (1.83 in the dry and 0.83 in the wet season). Overall, forest and utility areas exhibited higher RAI values compared to agriculture, with rodent abundance generally higher in the dry season, particularly in forested areas. The RAI remained relatively stable between seasons in utility areas at Wiang Kaen, while agriculture consistently showed the lowest rodent abundance across all sites and seasons. These findings highlight seasonal and habitat-dependent variations in rodent abundance, with a tendency for higher RAI in forests and utility areas, especially during the dry season.

Province and	Rat data					
Season	Family	Genus	Species	No. of Animal		
				Dry	Wet	Total
Chantaburi	Muridae	Rattus	Rattus exulans	30	32	
			Rattus rattus complex	29	24	
		Mus	Mus cervicolor	4	2	
			Mus caroli	2	0	
			Mus sp.	1	0	
		Maxomys	Maxomys surifer	3	1	
		Berylmys	Berylmys berdmorei	1	0	
		Bandicota	Bandicota indica	0	1	
	Total			70	60	130
Chiang Rai	Muridae	Rattus	Rattus exulans	60	37	
			Rattus rattus complex	23	27	
			Rattus andamanensis	15	0	
		Mus	Mus pahari	1	0	
		Maxomys	Maxomys surifer	0	1	
		Berylmys	Berylmys berdmorei	3	1	
			Berylmys bowersi	0	1	
		Niviventer	Niviventer fulvescens	0	3	
		Hapalomys	Hapalomys delacouri	0	1	
		Leopoldamys	Leopoldamys edwardsi	0	1	
	Total			102	72	174
Summary				172	132	304

For rat biological sample collections, a total of 304 rats were investigated across two provinces. The samples obtained included 296 oral swabs, 284 rectal swabs, 259 blood samples, 132 urine samples, and 136 fecal samples. These samples were systematically collected to ensure comprehensive pathogen detection and analysis. The detailed distribution of biological sample types across different rat species and provinces is presented in Table 5.3.

Table 4.3.1 Species composition of rodents collection in Chanthaburi and Chaing Rai provinces in 2024.

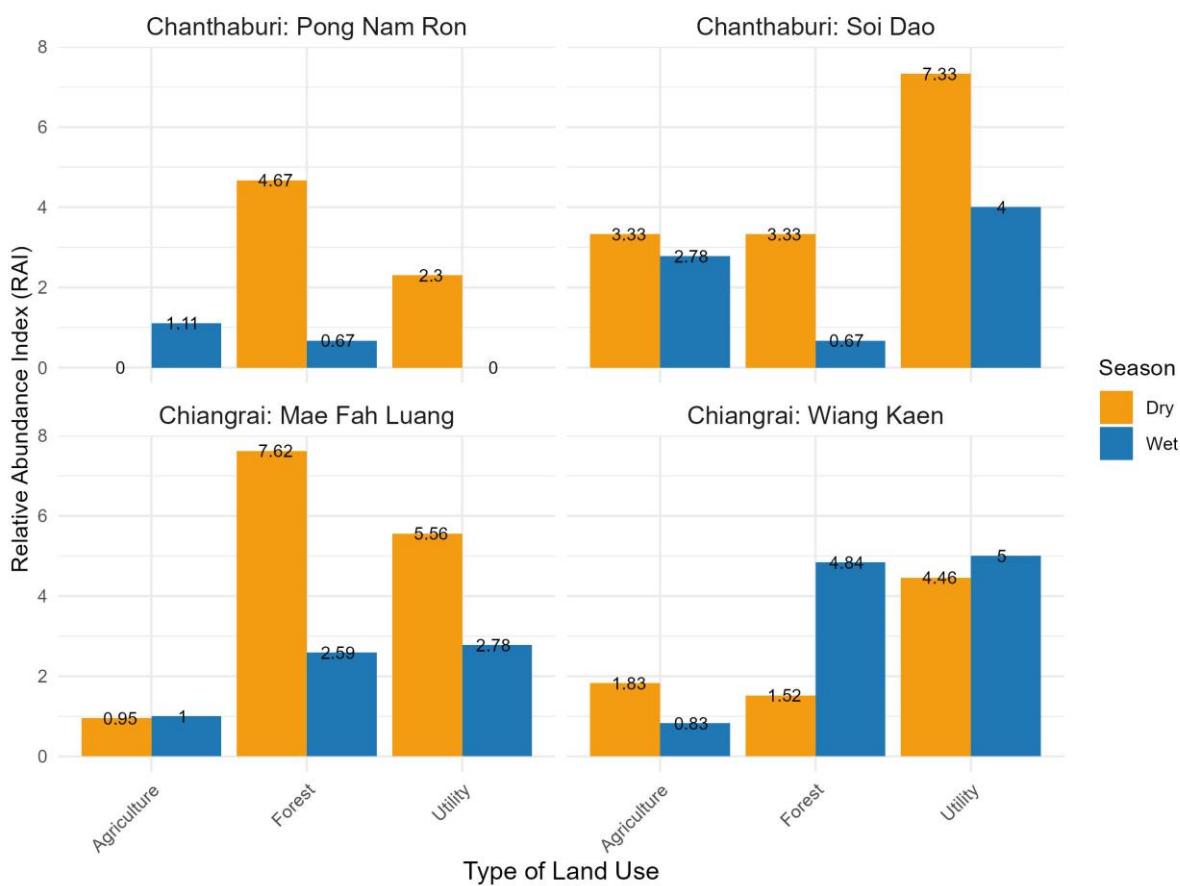


Figure 5.3-1 Relative Abundance Index (RAI) of rats across study sites in Chanthaburi and Chaing Rai in 2024

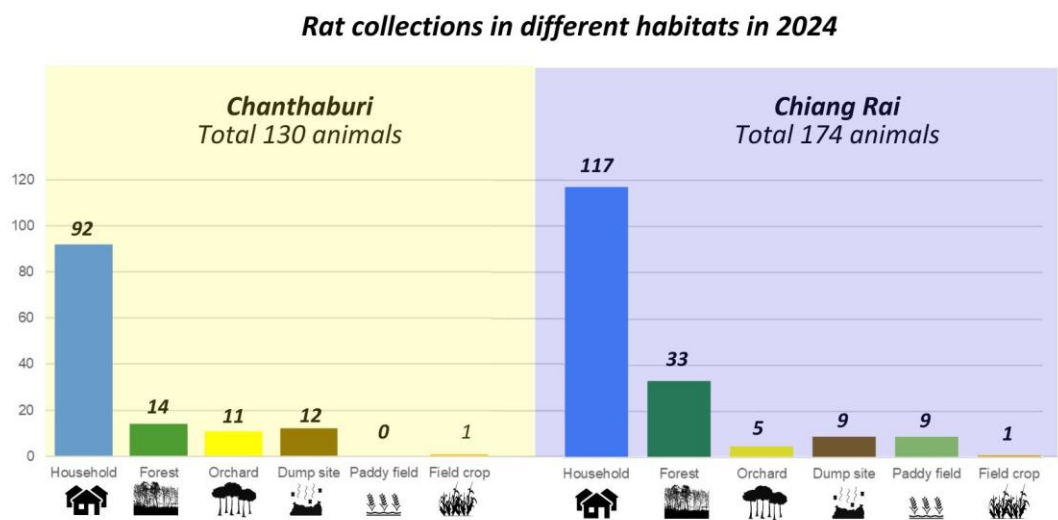


Figure 5.3-2 Rat collections in different six habitats in this study from Chanthaburi and Chiang Rai in 2024

Table 5.3-1 The rat species diversity in different habitats across Chanthaburi and Chiang Rai provinces in 2024.

Habitat	Species	No. of animal
Household	<i>Rattus exulans</i>	156
	<i>Rattus rattus complex</i>	48
	<i>Berylmys berdmorei</i>	3
	<i>Bandicota indica</i>	1
	<i>Mus caroli</i>	1
		209
Forest	<i>Rattus rattus complex</i>	22
	<i>Rattus andamanensis</i>	13
	<i>Maxomys surifer</i>	5
	<i>Niviventer fulvescens</i>	3
	<i>Hapalomys delacouri</i>	1
	<i>Leopoldamys edwardsi</i>	1
	<i>Mus pahari</i>	1
	<i>Rattus exulans</i>	1
		47
Dumpsite	<i>Rattus rattus complex</i>	13
	<i>Mus cervicolor</i>	4
	<i>Rattus exulans</i>	2
	<i>Berylmys berdmorei</i>	1
	<i>Mussp.</i>	1
		21
Orchard	<i>Rattus rattus complex</i>	11
	<i>Rattus andamanensis</i>	2
	<i>Berylmys bowersi</i>	1
	<i>Mus caroli</i>	1
	<i>Mus cervicolor</i>	1
		16
Paddy field	<i>Berylmys berdmorei</i>	1
	<i>Rattus rattus complex</i>	8
		9
Field crop	<i>Mus cervicolor</i>	1
	<i>Rattus rattus complex</i>	1
		2
Total		304

Table 5.3-2 Biological sample collection from rats in Chanthaburi and Chaing Rai in 2024.

Province	Species	No. of animal	Oral swab	Rectal swab	Blood	Urine	Feces
Chanthaburi	<i>Rattus exulans</i>	62	62	60	49	32	32
	<i>Rattus rattus complex</i>	53	53	53	40	24	25
	<i>Mus cervicolor</i>	6	6	4	5	2	2
	<i>Maxomys surifer</i>	4	4	4	3	1	1
	<i>Mus caroli</i>	2	2	0	2	0	0
	<i>Bandicota indica</i>	1	1	1	1	1	1
	<i>Berylmys berdmorei</i>	1	1	1	0	0	0
	<i>Mussp.</i>	1	1	1	1	0	0
		130	130	124	101	60	61
Chiang Rai	<i>Rattus exulans</i>	97	90	87	92	37	38
	<i>Rattus rattus complex</i>	50	49	46	44	27	28
	<i>Rattus andamanensis</i>	15	15	15	12	0	1
	<i>Berylmys berdmorei</i>	4	4	4	4	1	1
	<i>Niviventer fulvescens</i>	3	3	3	2	3	3
	<i>Berylmys bowersi</i>	1	1	1	1	1	1
	<i>Hapalomys delacouri</i>	1	1	1	0	1	1
	<i>Leopoldamys edwardsi</i>	1	1	1	1	1	1
	<i>Maxomys surifer</i>	1	1	1	1	1	1
	<i>Mus pahari</i>	1	1	1	1	0	0
		174	166	160	158	72	75
Total		304	296	284	259	132	136

5.4. Secondary wildlife

A total of 24 secondary wildlife animals were investigated, with 12 animals from Chanthaburi and 12 from Chiang Rai. Across both provinces, the species captured included 9 *Menetes berdmorei*, 4 *Tupaia belangeri*, 4 birds of different species, 2 *Sciurus aureogaster*, and one each of *Callosciurus finlaysonii*, *Hylomys peguensis*, *Hystrix brachyura*, *Paradoxurus hermaphroditus*, and *Prionodon pardicolor*.

For habitat analysis, the secondary wildlife animals were obtained from six distinct habitats: forest, agro-forest, orchard, dumpsite, household, and market. The majority of animals were obtained from the orchard habitat, with 7 individuals, followed by 5 animals from the forest,

4 animals from the market, and 2 animals from each agro-forest, dumpsite, and household. Detailed data on the habitat distribution of secondary wildlife are provided in Table 5.4.1. From the 24 secondary wildlife animals, a total of 24 oral swabs, 6 nasal swabs, 5 blood samples, 5 organ tissue samples, and 3 fecal samples were collected. Organ tissue specimens, which included lung, spleen, kidney, liver, intestine, and heart, were obtained from animals purchased at a bush meat market in Chiang Rai. The number of biological specimens collected from each animal species in both Chanthaburi and Chiang Rai is detailed in Table 5.4.2.

Table 5.4.-1 Secondary wildlife species collected in different habitats in Chanthaburi and Chiang Rai provinces in 2024.

Province	Species	Habitat					
		Forest	Agro-forest	Orchard	Dumpsite	Household	Market
Chanthaburi	<i>Black-winged pitta</i>			1			
	<i>Common tailorbird</i>			1			
	<i>Greater racket-tailed drongo</i>			1			
	<i>Menetes berdmorei</i>	1		3	2		
	<i>Oriental magpie-robin</i>			1			
	<i>Tupaia belangeri</i>	2					
		3		7	2	0	12
Chiang Rai	<i>Asian palm civet</i>						1
	<i>Finlayson's Squirrel</i>						1
	<i>Hylomys peguensis</i>	1					
	<i>Malayan Porcupine</i>						1
	<i>Menetes berdmorei</i>	1				2	
	<i>Red-bellied squirrel</i>		2				
	<i>Spotted linsang</i>						1
	<i>Tupaia belangeri</i>				1	1	
		2	2	0	1	3	4
							12
Summary		5	2	7	2	2	4
							24

Table 5.4-2 Table 4.4.2 Biological sample collection from secondary wildlife in Chanthaburi and Chaing Rai in 2024.

Province	Species	No. of animal	Nasal swab	Oral swab	Blood	Feces	Organ
Chanthaburi	<i>Black-winged pitta</i>	1	0	1	0	0	0
	<i>Common tailorbird</i>	1	0	1	0	0	0
	<i>Greater racket-tailed drongo</i>	1	0	1	0	0	0
	<i>Menetes berdmorei</i>	6	0	6	1	1	0
	<i>Oriental magpie-robin</i>	1	0	1	0	0	0
	<i>Tupaia belangeri</i>	2	0	2	0	0	0
		12	0	12	1	1	0
Chiang Rai	<i>Asian palm civet</i>	1	1	1	0	0	1
	<i>Finlayson's Squirrel</i>	1	1	1	0	0	1
	<i>Hylomys peguensis</i>	1	0	1	1	1	0
	<i>Malayan Porcupine</i>	1	1	1	0	0	0
	<i>Menetes berdmorei</i>	3	0	3	2	1	0
	<i>Red-bellied squirrel</i>	2	2	2	0	0	2
	<i>Spotted linsang</i>	1	1	1	0	0	1
	<i>Tupaia belangeri</i>	2	0	2	1	0	0
		12	6	12	4	2	5
Grand Total		24	6	24	5	3	5

5.5. Domestic animals

In this study, a total of 316 domestic animals were investigated, including 190 from Chanthaburi and 126 from Chiang Rai. The investigated domestic animals comprised various species, with the majority being domestic dogs (*Canis lupus*, 203 individuals) and domestic cats (*Felis catus*, 108 individuals). Additionally, three pigs (*Sus scrofa*), one Muscovy duck (*Anas platyrhynchos*), and one rabbit (*Oryctolagus cuniculus*) were examined. The distribution of domestic animal species across different seasons and provinces is illustrated in Figure 5.5.1.

Domestic animals in this study were obtained from three distinct habitats: Village, Temple, and Forest Edge. The majority of the animals, accounting for 91.45% (278 out of 304), were collected from the Village habitat. A smaller number of animals were captured from the Forest Edge (23 animals) and the Temple (15 animals). The distribution of domestic animals across these habitats in each province is illustrated in Figure 5.5.2.

In this study, the living types of domestic animals were analysed to assess potential risks of pathogen exposure. Domestic animals were categorized into three living types: free-roaming, semi-indoor, and indoor. The majority of the animals, 74.01% (225 out of 304), were classified as free roaming, followed by 76 semi-indoor animals and 15 indoor animals. The distribution of domestic animals across these living types in each province is presented in Figure 5.5.3.

For biological sample collection aimed at pathogen detection, a total of 35 nasal swabs, 313 oral swabs, 299 rectal swabs, and 254 blood samples were obtained from all 316 domestic animals. The lower number of nasal swabs was attributed to the limitations of physical restraint techniques used during sample collection from domestic animals. The detailed distribution of biological samples collected is provided in Table 4.5.1.

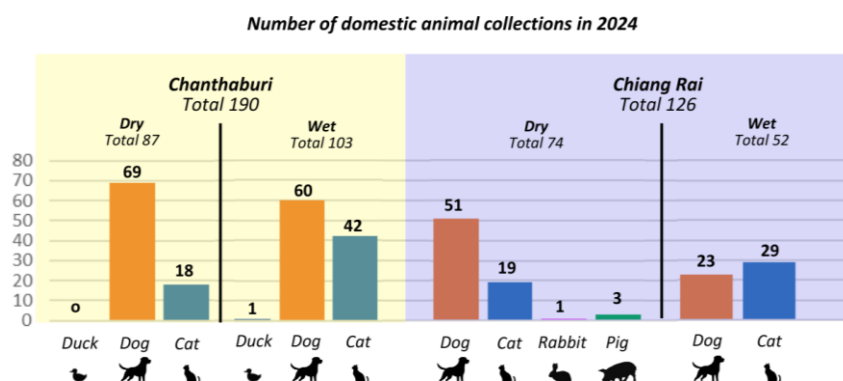


Figure 5.5-1 Figure 4.5.1 The information of investigated domestic animals from Chanthaburi and Chiang Rai in 2024.

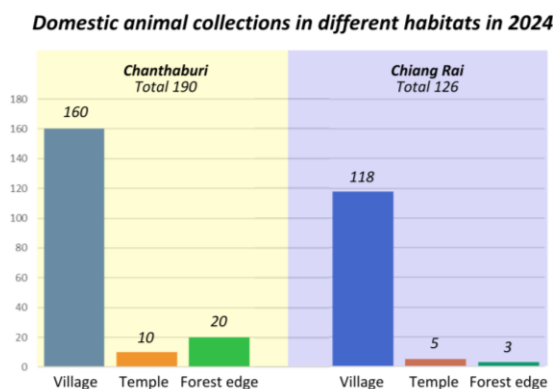


Figure 5.5-2 The investigated domestic animal collections in different three habitats in this study

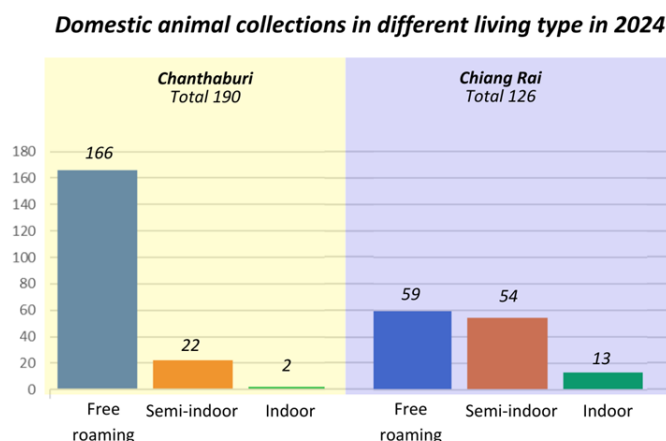


Figure 5.5-3 The investigated domestic animal collections in different three living types in this study

Table 5.5-1 Table 4.5.1 Biological sample collection from domestic animals in Chanthaburi and Chaing Rai in 2024.

Province	Species	No. of Animal	Nasal swab	Oral swab	Rectal swab	Blood
Chanthaburi	<i>Canis lupus</i>	129	5	127	122	120
	<i>Felis catus</i>	60	1	60	55	30
	<i>Anas platyrhynchos</i>	1	0	1	1	1
		190	6	188	178	151
Chiang Rai	<i>Canis lupus</i>	74	8	74	73	69
	<i>Felis catus</i>	48	18	47	47	31
	<i>Sus scrofa</i>	3	3	3	0	3
	<i>Oryctolagus cuniculus</i>	1	0	1	1	0
		126	29	125	121	103
Summary		316	35	313	299	254

6. Discussion

In this study, a total of 1,056 animals were collected across two provinces, including bats, rats, secondary wildlife, and domestic animals. Bats were the most numerous, with 412 individuals representing multiple species and diverse habitats, such as caves, forests, and agricultural areas. A total of 304 rats were captured, with higher species diversity observed in forested habitats compared to human settlements. Secondary wildlife, including squirrels, tree shrews, civets, and porcupines, were collected from various habitats, with orchards being the most frequent source. Additionally, 316 domestic animals, primarily dogs and cats, were investigated, with the majority found in villages. The variety of habitats and species collected highlights the ecological complexity of wildlife-human interactions and underscores the importance of monitoring these animal groups for potential zoonotic pathogens.

Biological samples were systematically collected from all animal groups to ensure comprehensive pathogen surveillance. For bats and rats, oral and rectal swabs were the primary samples, along with urine, feces, and blood, to maximize pathogen detection. In secondary wildlife, a combination of swabs, blood, and organ tissues was obtained, particularly from carcasses purchased at bush meat markets, providing a unique opportunity to investigate pathogen reservoirs. Similarly, for domestic animals, nasal, oral, and rectal swabs were collected, though nasal swabs were more limited due to physical restraint challenges. The diverse range of biological specimens obtained across species enhances the likelihood of detecting potential zoonotic agents and contributes to a more holistic understanding of pathogen circulation in wildlife and domestic animals.

The next step of this study would focus on pathogen detection, aiming to identify viral, bacterial, and parasitic agents present in the collected biological samples. Given the diversity of animals and their habitats, this investigation will provide insights into potential spillover risks from wildlife to domestic animals and humans. Special attention will be given to species with high human-wildlife interface, such as bats and rodents, which are known reservoirs of zoonotic pathogens. The findings will contribute to risk assessments for zoonotic spillover and pandemic emergence, informing public health and conservation strategies. Future studies

should integrate genetic analysis, ecological modelling, and serological surveys to further elucidate the mechanisms driving pathogen transmission and develop targeted interventions to mitigate future outbreaks.

7. Reference

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